MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCIAM100 INTERACTIVE ROBOTIC SYSTEMS

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	What is the microcontroller used in the Arduino Board A) ATmega328p B) ATmega2560 C) ATmega32U4 D) ATmega8	R	1
2	The Speed of sound in air is------ A) 3m/s B) 3000m/s C) 34m/s D)343m/s	R	2
3	Which is the communication protocol used by the MPU6050 to communicate with microcontrollers. A) UART B) SPI C) I2C D) CAN	R	3
4	---------- Function is called repeatedly in an Arduino program. A) void setup() B) void loop() C) main() D) Init()	R	1
5	Which Component is NOT typically found in an ultrasonic sensor? A) Transmitter B) Receiver C) Amplifier D) Solar panel	U	2
6	How many number of pins does the MPU6050 sensor typically have? A) 4 B) 6 C) 8 D) 10	R	3
7	The correct file extension for the Arduino IDE installation file is A) .exe B) .zip C) .tar.gz D) .dmg	U	1
8	The principle of operation of MQ 2 smoke sensor is the change of _____ in the presence of gas. A) voltage B) Resistance C) Current D) Capacitance	U	2
9	Which model is commonly used to represent self-balancing robots? A) Simple Pendulum Model B) Inverted Pendulum Model C) Circular Motion Model D) Rotational Model	U	3
10	What is the typical output range of a standard servo motor when controlled via PWM? A) 0-90 degrees B) 0-180 degrees C) 0-360 degrees D) 0-45 degrees	R	3

1 * 10 = 10

Part B

Short Answer Questions

Answer **4** Questions

Each question carries 5 marks

11	Explain the difference between the void setup () and void loop () in Arduino? Give an example?	A	1
12	Describe the working principle of ultrasonic sensor in detail?	U	2
13	Explain how Arduino platform contributed to the growth of the Internet of Things (IoT)?	U	1
14	Explain the function of an ultrasonic sensor in RADAR mode?	U	3
15	Explain the working principle of IR Flame sensor?	U	2
16	What is the primary function of a self-balancing robotic system? Explain in detail.	U	3

4 * 5 = 20

Part C

Essay Questions

Answer **2** Questions

Each question carries 10 marks

17	Compare void setup () and void loop () functions in Arduino programming. How do they work together to create a complete Arduino Sketch? And explain the uses of both functions?	U	1
18	Write a program to read the values from LDR with proper steps?	A	2
19	Describe the fundamental principles that enable self-balancing robots to maintain stability. What technologies and mechanisms do these robots employ to achieve balance?	U	3
20	Explain the fundamental principles behind servo motors. What are the main components of a servo system, and how do they work together to achieve precise control of movement?	U	2

2 * 10 = 20